AMCS1043 Database Development and Applications Assignment Submission Form

Programme	DCS1S2
Tutorial Group	Group 7
Lecturer / Tutor	PHAIK LING ONG
Submission Date	20/12/2025

No.	Student Name (Please fill in the names in ascending order)	Student ID	Signature	Contribution (%)	Marks
1.	Eugene Chong Jin Le	25WMD07110		20	
2.	Ram	25WMD07438		20	
3.	Samuel Ho Dao Ern	25WMD07155		20	
4.	Wong Jun Hoe	25WMD07156		20	
5.	Wong Yi Hoe	25WMD07160		20	
				100	

Feedback / Comments:

AMCS1043 Database Development and Applications

DCS1S2

Year 1

Group 7

Plagiarism Statement

Read, complete, and sign this statement to be submitted with the written report.

We confirm that the submitted work is all our own work and is in our own words.

Name (Block Capitals)	Registration No.	Signature
1. RAM	25WMD07438	
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AMCS1043 Database Development and Applications

Assignment Assessment Rubrics

Programme:_____ Tutorial Group : _____

Criteria	Marks	Excellent	Good	Average	Poor	Score
GROUP						
(1) Organisation background	10	Very high level organisational analysis and research, covering all relevant details. Very clear and excellent description of problem statement and solution. 9 – 10	High level organisational analysis and research, covering mostly relevant details. Good and clear description of problem statement and solution. 6 – 8	Some organisational analysis and research, barely covering relevant details. Acceptable and clear description of problem statement and solution. 3 – 5	Lack of organisational analysis and research, covering details mostly not relevant. Insufficient and unclear description of problem statement and solution. 0 – 2	
(2) Business rules	10	Excellent business	Good business rules. 6 – 8	Acceptable business	Poor or incorrect business rules. 0 – 2	

		rules. 9 – 10		rules. 3 – 5		
(3) Database Design (ERD)	10	Excellent database design. 9 – 10	Good database design. 6 – 8	Acceptable database design. 3 – 5	Poor database design. 0 – 2	
(4) Create database tables in Oracle	20	Excellent design of Oracle database with all requirements met. 17 – 20	Good design of Oracle database with major requirements met. 11 – 16	Acceptable design of Oracle database with some requirements met. 5 – 10	Poor design of Oracle database and does not meet most of the requirements shown. 0 – 4	
(5) Create records	20	Excellent and sufficient records resemble real world data.	Good and sufficient records resemble real world data.	Acceptable and some adequate records resemble real world data.	Poor and unrealistic data.	

		17 – 20	11 – 16	5 – 10	0 – 4	
(7) Assignment report	10	Excellent writing style. Messages and answers are clear and effective. Full compliance of format. 9 – 10	Good writing style. Messages and answers sometimes lack clarity. Close compliance of format. 6 – 8	Average writing style. Messages and answers are poorly conveyed. Minimum compliance of format. 3 – 5	Writing style is poor. Answers are poorly conveyed. Non-compliance of format. 0 – 2	
Total	80					
INDIVIDUAL						
(6) Create query	10	Excellent and correct design of query statement based on the purpose/ scenario with excellent formatting. 9 – 10	Good and correct design of query statement based on the purpose/ scenario with appropriate formatting. 6 – 8	Average and correct design query statement based on the purpose/ scenario with appropriate formatting 3 – 5	Poor and incorrect design query statement with poor formatting. 0 – 2	

(8) Presentation	10	Excellent communication skills. Messages and answers are clear and effective. 9 – 10	Good communication skills. Messages and answers sometimes lack clarity. 6 – 8	Average communication skill. Messages and answers are poorly conveyed. 3 – 5	Use of language is ineffective; communication is generally poor. Answers are poorly conveyed. 0 – 2	
Total	20					

Appendix 3

AMCS1043 Database Development and Applications

Name (Please fill in the names in ascending order)	Query (10%)	Presentation (10%)	Marks (Task 6 & 8)	Remark (if any)	
1.					
2.					
3.					
4.					
5.					
6.					

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1.0 Organisation Background

Organisation Name: MediLife Family Clinic

1.1 Business Nature:

MediLife Family Clinic is a private outpatient clinic that provides general medical services. The clinic focuses on family healthcare, children's treatment, simple medical procedures, and basic check-ups.

1.2 Business Operations:

The clinic opens daily from 8:00 AM to 10:00 PM. Its daily workflow includes:

1. Patient can go directly to the clinic without an appointment, or register in advance

2. Doctor consultation

3. Pharmacy medication dispensing

4. Billing and payment

These operations involve a lot of data recording, such as patient details, consultation notes, medical history, appointment schedules, and medicine stock updates.

1.3 Products & Services:

- Services: Medical consultation, health screening, blood tests, vaccination, and minor wound dressing.
- Products: Medications such as antibiotics, cough medicine, supplements, and basic medical supplies like bandages.

1.4 Problem Statement

Currently, the clinic uses manual data recording methods such as paper files and Excel sheets. These methods create several problems in data management.

Problem 1 – Slow Patient Data Retrieval

Patient records are kept inside physical folders. When a returning patient comes, staff need 10–15 minutes to search for their record. This causes long queues and delays because the clinic has no digital system to store and retrieve patient data quickly.

Problem 2 – Poor Inventory Tracking

Medication stock is not recorded properly. There is no database table to track stock quantity, expiry dates, or items dispensed. This leads to:

- Running out of important medicines
- Keeping expired items
- Financial losses caused by poor inventory control

Problem 3 – Manual Billing Errors

Billing is done manually using calculators and paper. Without a proper billing database or transaction system, staff sometimes:

- Incorrect charge amount
- Miss medication charges
- Record payments incorrectly

This affects the clinic's financial accuracy.

1.5 & 1.6 Goals and Objectives

The goal is to design a relational database system that helps the clinic store, manage, and retrieve data more efficiently.

Objectives:

1. To create a database that stores all patient information, consultation history, and appointments in a central system for fast data retrieval.

2. To build an inventory module that automatically updates medicine stock levels whenever an item is dispensed.

3. To automate the billing process so that service fees, medication prices, and total charges are calculated accurately inside the database.

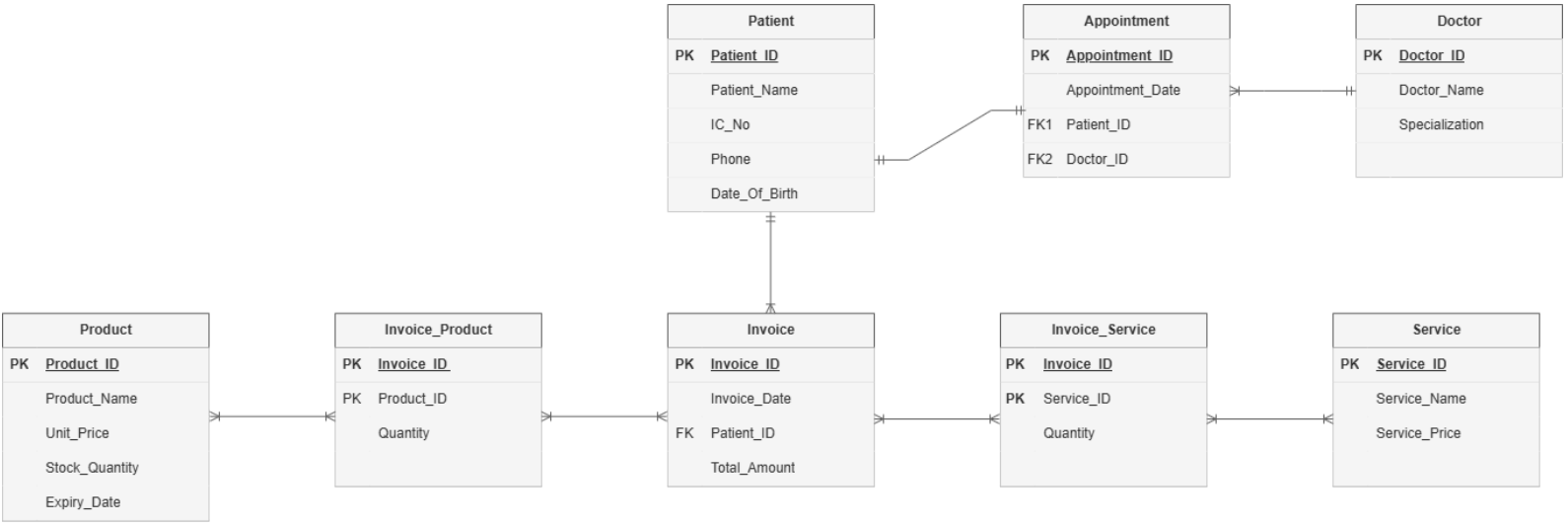
2.0 Business Rules

1. **Each appointment booked must be associated with exactly one patient and one doctor**.
2. An invoice **must belong to exactly one patient**, and a patient **may have multiple invoices** over time.
3. An invoice **may include multiple services**, and a service **may appear on multiple invoices**.

4. An invoice **may include multiple products**, and a product **may be sold in multiple invoices**.
5. A product or service **cannot be added to an invoice unless it exists in the PRODUCT or SERVICE table**, and duplicate product–invoice or service–invoice combinations **are not allowed**.

3.0 Database Design

3.1Entity-Relationship Diagram



3.2 List of all attributes in Database Design Language format

PRODUCT (Product_ID, Product_Name, Unit_Price, Stock_Quantity, Expiry_Date)

Invoice_Product (Invoice_ID, Product_ID, Quantity)

Invoice (Invoice_ID, Invoice_Date, Patient_ID*, Total_Amount)

Patient (Patient_ID, Patient_Name, IC_No, Phone, Date_Of_Birth)

Invoice_Service (Invoice_ID, Service_ID, Quantity)

Appointment (Appointment_ID, Appointment_Date, Patient_ID*, Doctor_ID*)

Service (Service_ID, Service_Name, Service_Price)

Doctor (Doctor_ID, Doctor_Name, Specialization)

4.0 Create Table

4.1 Create Product Table

```
SQL> CREATE TABLE PRODUCT (
  2       Product_ID          NUMBER PRIMARY KEY,
  3       Product_Name        VARCHAR2(100) NOT NULL,
  4       Unit_Price          NUMBER(8,2) NOT NULL,
  5       Stock_Quantity      NUMBER NOT NULL,
  6       Expiry_Date         DATE
  7  );
CREATE TABLE PRODUCT (
```

4.2 Create Invoice Product Table

```
SQL> CREATE TABLE INVOICE_PRODUCT (
  2      Invoice_ID         NUMBER NOT NULL,
  3      Product_ID         NUMBER NOT NULL,
  4      Quantity           NUMBER NOT NULL,
  5      CONSTRAINT pk_invoice_product
  6          PRIMARY KEY (Invoice_ID, Product_ID),
  7      CONSTRAINT fk_ip_invoice
  8          FOREIGN KEY (Invoice_ID) REFERENCES INVOICE(Invoice_ID),
  9      CONSTRAINT fk_ip_product
 10          FOREIGN KEY (Product_ID) REFERENCES PRODUCT(Product_ID)
 11  );
```

4.3 Create Invoice Table

```
SQL> CREATE TABLE INVOICE (
  2      Invoice_ID         NUMBER PRIMARY KEY,
  3      Invoice_Date       DATE NOT NULL,
  4      Patient_ID         NUMBER NOT NULL,
  5      Total_Amount       NUMBER(10,2),
  6      CONSTRAINT fk_invoice_patient
  7          FOREIGN KEY (Patient_ID) REFERENCES PATIENT(Patient_ID)
  8  );
```

4.4 Create Patient Table

```
SQL> CREATE TABLE PATIENT (
  2      Patient_ID        NUMBER PRIMARY KEY,
  3      Patient_Name      VARCHAR2(100) NOT NULL,
  4      IC_No             VARCHAR2(20) UNIQUE NOT NULL,
  5      Phone             VARCHAR2(20),
  6      Date_Of_Birth     DATE
  7  );
```

4.5 Create Invoice Service Table

```
SQL> CREATE TABLE INVOICE_SERVICE (
  2      Invoice_ID        NUMBER NOT NULL,
  3      Service_ID        NUMBER NOT NULL,
  4      Quantity          NUMBER DEFAULT 1 NOT NULL,
  5      CONSTRAINT pk_invoice_service
  6          PRIMARY KEY (Invoice_ID, Service_ID),
  7      CONSTRAINT fk_is_invoice
  8          FOREIGN KEY (Invoice_ID) REFERENCES INVOICE(Invoice_ID),
  9      CONSTRAINT fk_is_service
 10          FOREIGN KEY (Service_ID) REFERENCES SERVICE(Service_ID)
 11  );
```

4.6 Create Appointment Table

```
SQL> CREATE TABLE APPOINTMENT (
  2      Appointment_ID    NUMBER PRIMARY KEY,
  3      Appointment_Date  DATE NOT NULL,
  4      Patient_ID        NUMBER NOT NULL,
  5      Doctor_ID         NUMBER NOT NULL,
  6      CONSTRAINT fk_appointment_patient
  7          FOREIGN KEY (Patient_ID) REFERENCES PATIENT(Patient_ID),
  8      CONSTRAINT fk_appointment_doctor
  9          FOREIGN KEY (Doctor_ID) REFERENCES DOCTOR(Doctor_ID)
 10  );
```

4.7 Create Service Table

```
SQL> CREATE TABLE SERVICE (
  2      Service_ID        NUMBER PRIMARY KEY,
  3      Service_Name      VARCHAR2(100) NOT NULL,
  4      Service_Price     NUMBER(8,2) NOT NULL
  5  );
```

4.8 Create Doctor Table

```
SQL> CREATE TABLE DOCTOR (
  2       Doctor_ID          NUMBER PRIMARY KEY,
  3       Doctor_Name        VARCHAR2(100) NOT NULL,
  4       Specialization     VARCHAR2(50)
  5  );
```

5.0 Create and Insert Records

5.1 Insert Product Records

```
INSERT INTO PRODUCT (Product_ID, Product_Name, Unit_Price, Stock_Quantity, Expiry_Date)
VALUES (1301, 'Paracetamol', 5.00, 500, DATE '2026-12-31');

INSERT INTO PRODUCT (Product_ID, Product_Name, Unit_Price, Stock_Quantity, Expiry_Date)
VALUES (1302, 'Amoxicillin', 12.00, 300, DATE '2026-10-31');

INSERT INTO PRODUCT (Product_ID, Product_Name, Unit_Price, Stock_Quantity, Expiry_Date)
VALUES (1303, 'Cough Syrup', 8.50, 200, DATE '2026-08-31');

INSERT INTO PRODUCT (Product_ID, Product_Name, Unit_Price, Stock_Quantity, Expiry_Date)
VALUES (1304, 'Vitamin C', 15.00, 400, DATE '2027-01-31');

INSERT INTO PRODUCT (Product_ID, Product_Name, Unit_Price, Stock_Quantity, Expiry_Date)
VALUES (1305, 'Ibuprofen', 7.50, 350, DATE '2026-11-30');

INSERT INTO PRODUCT (Product_ID, Product_Name, Unit_Price, Stock_Quantity, Expiry_Date)
VALUES (1306, 'Antacid', 6.00, 250, DATE '2026-09-30');

INSERT INTO PRODUCT (Product_ID, Product_Name, Unit_Price, Stock_Quantity, Expiry_Date)
VALUES (1307, 'Insulin', 90.00, 100, DATE '2025-12-31');

INSERT INTO PRODUCT (Product_ID, Product_Name, Unit_Price, Stock_Quantity, Expiry_Date)
VALUES (1308, 'Eye Drops', 18.00, 180, DATE '2026-07-31');

INSERT INTO PRODUCT (Product_ID, Product_Name, Unit_Price, Stock_Quantity, Expiry_Date)
VALUES (1309, 'Bandage', 3.00, 600, NULL);

INSERT INTO PRODUCT (Product_ID, Product_Name, Unit_Price, Stock_Quantity, Expiry_Date)
VALUES (1310, 'Glucose Drink', 4.50, 500, DATE '2026-06-30');
```

5.2 Insert Invoice Product Table

```
INSERT INTO INVOICE_PRODUCT (Invoice_ID, Product_ID, Quantity)
VALUES (1501, 1301, 10);

INSERT INTO INVOICE_PRODUCT (Invoice_ID, Product_ID, Quantity)
VALUES (1501, 1304, 2);

INSERT INTO INVOICE_PRODUCT (Invoice_ID, Product_ID, Quantity)
VALUES (1502, 1302, 5);

INSERT INTO INVOICE_PRODUCT (Invoice_ID, Product_ID, Quantity)
VALUES (1503, 1303, 1);

INSERT INTO INVOICE_PRODUCT (Invoice_ID, Product_ID, Quantity)
VALUES (1504, 1307, 1);

INSERT INTO INVOICE_PRODUCT (Invoice_ID, Product_ID, Quantity)
VALUES (1505, 1305, 3);

INSERT INTO INVOICE_PRODUCT (Invoice_ID, Product_ID, Quantity)
VALUES (1506, 1306, 4);

INSERT INTO INVOICE_PRODUCT (Invoice_ID, Product_ID, Quantity)
VALUES (1507, 1308, 1);

INSERT INTO INVOICE_PRODUCT (Invoice_ID, Product_ID, Quantity)
VALUES (1508, 1309, 10);

INSERT INTO INVOICE_PRODUCT (Invoice_ID, Product_ID, Quantity)
VALUES (1509, 1310, 2);
```

5.3 Insert Invoice Table

```
-- INVOICE
INSERT INTO INVOICE (Invoice_ID, Invoice_Date, Patient_ID, Total_Amount)
VALUES (1501, SYSDATE-20, 1001, 200);
INSERT INTO INVOICE (Invoice_ID, Invoice_Date, Patient_ID, Total_Amount)
VALUES (1502, SYSDATE-19, 1002, 350);
INSERT INTO INVOICE (Invoice_ID, Invoice_Date, Patient_ID, Total_Amount)
VALUES (1503, SYSDATE-18, 1003, 120);
INSERT INTO INVOICE (Invoice_ID, Invoice_Date, Patient_ID, Total_Amount)
VALUES (1504, SYSDATE-17, 1004, 500);
INSERT INTO INVOICE (Invoice_ID, Invoice_Date, Patient_ID, Total_Amount)
VALUES (1505, SYSDATE-16, 1005, 90);
INSERT INTO INVOICE (Invoice_ID, Invoice_Date, Patient_ID, Total_Amount)
VALUES (1506, SYSDATE-15, 1006, 180);
INSERT INTO INVOICE (Invoice_ID, Invoice_Date, Patient_ID, Total_Amount)
VALUES (1507, SYSDATE-14, 1007, 260);
INSERT INTO INVOICE (Invoice_ID, Invoice_Date, Patient_ID, Total_Amount)
VALUES (1508, SYSDATE-13, 1008, 400);
INSERT INTO INVOICE (Invoice_ID, Invoice_Date, Patient_ID, Total_Amount)
VALUES (1509, SYSDATE-12, 1009, 150);
INSERT INTO INVOICE (Invoice_ID, Invoice_Date, Patient_ID, Total_Amount)
VALUES (1510, SYSDATE-11, 1010, 300);
```

5.4 Insert Patient Table

```
-- PATIENT
INSERT INTO PATIENT (Patient_ID, Patient_Name, IC_No, Phone, Date_Of_Birth)
VALUES (1001, 'Ali Hassan', '900101-01-1111', '0123456789', DATE '1990-01-01');
INSERT INTO PATIENT (Patient_ID, Patient_Name, IC_No, Phone, Date_Of_Birth)
VALUES (1002, 'Siti Aminah', '920202-02-2222', '0134567890', DATE '1992-02-02');
INSERT INTO PATIENT (Patient_ID, Patient_Name, IC_No, Phone, Date_Of_Birth)
VALUES (1003, 'John Tan', '880303-03-3333', '0145678901', DATE '1988-03-03');
INSERT INTO PATIENT (Patient_ID, Patient_Name, IC_No, Phone, Date_Of_Birth)
VALUES (1004, 'Nur Aisyah', '950404-04-4444', '0156789012', DATE '1995-04-04');
INSERT INTO PATIENT (Patient_ID, Patient_Name, IC_No, Phone, Date_Of_Birth)
VALUES (1005, 'Ahmad Zaki', '870505-05-5555', '0167890123', DATE '1987-05-05');
INSERT INTO PATIENT (Patient_ID, Patient_Name, IC_No, Phone, Date_Of_Birth)
VALUES (1006, 'Lim Wei', '910606-06-6666', '0178901234', DATE '1991-06-06');
INSERT INTO PATIENT (Patient_ID, Patient_Name, IC_No, Phone, Date_Of_Birth)
VALUES (1007, 'Aina Farah', '930707-07-7777', '0189012345', DATE '1993-07-07');
INSERT INTO PATIENT (Patient_ID, Patient_Name, IC_No, Phone, Date_Of_Birth)
VALUES (1008, 'Daniel Wong', '850808-08-8888', '0190123456', DATE '1985-08-08');
INSERT INTO PATIENT (Patient_ID, Patient_Name, IC_No, Phone, Date_Of_Birth)
VALUES (1009, 'Nur Syafiqah', '960909-09-9999', '0111234567', DATE '1996-09-09');
INSERT INTO PATIENT (Patient_ID, Patient_Name, IC_No, Phone, Date_Of_Birth)
VALUES (1010, 'Hafiz Rahman', '890101-10-1010', '0122345678', DATE '1989-01-10');
```

5.5 Insert Invoice Service Table

```
-- INVOICE_SERVICE
INSERT INTO INVOICE_SERVICE (Invoice_ID, Service_ID, Quantity)
VALUES (1501, 1201, 1);
INSERT INTO INVOICE_SERVICE (Invoice_ID, Service_ID, Quantity)
VALUES (1501, 1203, 1);
INSERT INTO INVOICE_SERVICE (Invoice_ID, Service_ID, Quantity)
VALUES (1502, 1204, 1);
INSERT INTO INVOICE_SERVICE (Invoice_ID, Service_ID, Quantity)
VALUES (1503, 1202, 1);
INSERT INTO INVOICE_SERVICE (Invoice_ID, Service_ID, Quantity)
VALUES (1504, 1206, 1);
INSERT INTO INVOICE_SERVICE (Invoice_ID, Service_ID, Quantity)
VALUES (1505, 1201, 1);
INSERT INTO INVOICE_SERVICE (Invoice_ID, Service_ID, Quantity)
VALUES (1506, 1205, 1);
INSERT INTO INVOICE_SERVICE (Invoice_ID, Service_ID, Quantity)
VALUES (1507, 1207, 1);
INSERT INTO INVOICE_SERVICE (Invoice_ID, Service_ID, Quantity)
VALUES (1508, 1208, 1);
INSERT INTO INVOICE_SERVICE (Invoice_ID, Service_ID, Quantity)
VALUES (1509, 1209, 1);
```

5.6 Insert Appointment Table

```
-- APPOINTMENT
INSERT INTO APPOINTMENT (Appointment_ID, Appointment_Date, Patient_ID, Doctor_ID)
VALUES (1401, SYSDATE-30, 1001, 1101);
INSERT INTO APPOINTMENT (Appointment_ID, Appointment_Date, Patient_ID, Doctor_ID)
VALUES (1402, SYSDATE-29, 1002, 1102);
INSERT INTO APPOINTMENT (Appointment_ID, Appointment_Date, Patient_ID, Doctor_ID)
VALUES (1403, SYSDATE-28, 1003, 1103);
INSERT INTO APPOINTMENT (Appointment_ID, Appointment_Date, Patient_ID, Doctor_ID)
VALUES (1404, SYSDATE-27, 1004, 1104);
INSERT INTO APPOINTMENT (Appointment_ID, Appointment_Date, Patient_ID, Doctor_ID)
VALUES (1405, SYSDATE-26, 1005, 1105);
INSERT INTO APPOINTMENT (Appointment_ID, Appointment_Date, Patient_ID, Doctor_ID)
VALUES (1406, SYSDATE-25, 1006, 1106);
INSERT INTO APPOINTMENT (Appointment_ID, Appointment_Date, Patient_ID, Doctor_ID)
VALUES (1407, SYSDATE-24, 1007, 1107);
INSERT INTO APPOINTMENT (Appointment_ID, Appointment_Date, Patient_ID, Doctor_ID)
VALUES (1408, SYSDATE-23, 1008, 1108);
INSERT INTO APPOINTMENT (Appointment_ID, Appointment_Date, Patient_ID, Doctor_ID)
VALUES (1409, SYSDATE-22, 1009, 1109);
INSERT INTO APPOINTMENT (Appointment_ID, Appointment_Date, Patient_ID, Doctor_ID)
VALUES (1410, SYSDATE-21, 1010, 1110);
```

5.7 Insert Service Table

```
-- SERVICE
INSERT INTO SERVICE (Service_ID, Service_Name, Service_Price)
VALUES (1201, 'Consultation', 50);
INSERT INTO SERVICE (Service_ID, Service_Name, Service_Price)
VALUES (1202, 'X-Ray', 120);
INSERT INTO SERVICE (Service_ID, Service_Name, Service_Price)
VALUES (1203, 'Blood Test', 80);
INSERT INTO SERVICE (Service_ID, Service_Name, Service_Price)
VALUES (1204, 'ECG', 150);
INSERT INTO SERVICE (Service_ID, Service_Name, Service_Price)
VALUES (1205, 'Ultrasound', 200);
INSERT INTO SERVICE (Service_ID, Service_Name, Service_Price)
VALUES (1206, 'MRI Scan', 800);
INSERT INTO SERVICE (Service_ID, Service_Name, Service_Price)
VALUES (1207, 'Physiotherapy', 100);
INSERT INTO SERVICE (Service_ID, Service_Name, Service_Price)
VALUES (1208, 'Vaccination', 60);
INSERT INTO SERVICE (Service_ID, Service_Name, Service_Price)
VALUES (1209, 'Dental Cleaning', 90);
INSERT INTO SERVICE (Service_ID, Service_Name, Service_Price)
VALUES (1210, 'Eye Checkup', 70);
```

5.8 Insert Doctor Table

```
-- DOCTOR
INSERT INTO DOCTOR (Doctor_ID, Doctor_Name, Specialization)
VALUES (1101, 'Dr. Ahmad', 'General');
INSERT INTO DOCTOR (Doctor_ID, Doctor_Name, Specialization)
VALUES (1102, 'Dr. Lim', 'Cardiology');
INSERT INTO DOCTOR (Doctor_ID, Doctor_Name, Specialization)
VALUES (1103, 'Dr. Wong', 'Dermatology');
INSERT INTO DOCTOR (Doctor_ID, Doctor_Name, Specialization)
VALUES (1104, 'Dr. Siti', 'Pediatrics');
INSERT INTO DOCTOR (Doctor_ID, Doctor_Name, Specialization)
VALUES (1105, 'Dr. Hafiz', 'Orthopedic');
INSERT INTO DOCTOR (Doctor_ID, Doctor_Name, Specialization)
VALUES (1106, 'Dr. Kumar', 'Neurology');
INSERT INTO DOCTOR (Doctor_ID, Doctor_Name, Specialization)
VALUES (1107, 'Dr. Aina', 'Gynecology');
INSERT INTO DOCTOR (Doctor_ID, Doctor_Name, Specialization)
VALUES (1108, 'Dr. John', 'ENT');
INSERT INTO DOCTOR (Doctor_ID, Doctor_Name, Specialization)
VALUES (1109, 'Dr. Sarah', 'Psychiatry');
INSERT INTO DOCTOR (Doctor_ID, Doctor_Name, Specialization)
VALUES (1110, 'Dr. Daniel', 'Ophthalmology');
```

6.0 Create Query

6.1 Total Healthcare Spending for a Patient (Samuel Ho Dao Ern)

```
SET VERIFY OFF
SET LINESIZE 120
SET PAGESIZE 60

SPOOL patient_receipt.txt

TTITLE CENTER 'PATIENT RECEIPT' SKIP 1
TTITLE CENTER 'Healthcare System' SKIP 2

ACCEPT p_patient_id PROMPT 'Enter Patient ID: '

-- ========================
-- Header: Patient Info
-- ========================
COLUMN patient_name   HEADING 'Patient Name' FORMAT A20
COLUMN invoice_id     HEADING 'Invoice ID' FORMAT 9999
COLUMN invoice_date   HEADING 'Invoice Date' FORMAT A12

SELECT
   p.patient_name,
   i.invoice_id,
   TO_CHAR(i.invoice_date,'YYYY-MM-DD') AS invoice_date
FROM patient p
JOIN invoice i ON p.patient_id = i.patient_id
WHERE p.patient_id = '&p_patient_id';

-- ========================
-- Services Section
-- ========================
```

```
COLUMN service_name HEADING 'Service' FORMAT A25
COLUMN service_qty  HEADING 'Qty' FORMAT 999
COLUMN service_amt  HEADING 'Amount (RM)' FORMAT 999,999.99

BREAK ON service_name SKIP 1
COMPUTE SUM LABEL 'Total Service Cost (RM):' OF service_amt ON REPORT

SELECT
    NVL(s.service_name,'-- No Services --') AS service_name,
    NVL(isv.quantity,0) AS service_qty,
    NVL((s.service_price * isv.quantity),0) AS service_amt
FROM invoice i
LEFT JOIN invoice_service isv ON i.invoice_id = isv.invoice_id
LEFT JOIN service s ON isv.service_id = s.service_id
WHERE i.patient_id = '&p_patient_id';

-- ========================
-- Medicines Section
-- ========================
COLUMN product_name HEADING 'Medicine' FORMAT A25
COLUMN product_qty  HEADING 'Qty' FORMAT 999
COLUMN product_amt  HEADING 'Amount (RM)' FORMAT 999,999.99

BREAK ON product_name SKIP 1
COMPUTE SUM LABEL 'Total Medicine Cost (RM):' OF product_amt ON REPORT

SELECT
    NVL(pr.product_name,'-- No Medicines --') AS product_name,
    NVL(ip.quantity,0) AS product_qty,
    NVL((pr.unit_price * ip.quantity),0) AS product_amt
FROM invoice i
LEFT JOIN invoice_product ip ON i.invoice_id = ip.invoice_id
```

```
LEFT JOIN product pr ON ip.product_id = pr.product_id
WHERE i.patient_id = '&p_patient_id';

-- ========================
-- Grand Total Section
-- ========================
COLUMN total_amt HEADING 'Grand Total (RM)' FORMAT 999,999.99

SELECT
    NVL(SUM(s.service_price * isv.quantity),0) +
    NVL(SUM(pr.unit_price * ip.quantity),0) AS total_amt
FROM invoice i
LEFT JOIN invoice_service isv ON i.invoice_id = isv.invoice_id
LEFT JOIN service s ON isv.service_id = s.service_id
LEFT JOIN invoice_product ip ON i.invoice_id = ip.invoice_id
LEFT JOIN product pr ON ip.product_id = pr.product_id
WHERE i.patient_id = '&p_patient_id';
SPOOL OFF;
```

```
SQL> start C:\Users\choon\Downloads\patient_medical_receipt.txt
Enter Patient ID: 1002

                                                    Healthcare System

Patient Name         Invoice ID Invoice Date
-------------------- ---------- ------------
Siti Aminah                1502 2025-11-30
Siti Aminah                1522 2025-11-10


                                                    Healthcare System

Service                    Qty Amount (RM)
------------------------- ---- -----------
Consultation                 1       50.00

ECG                          1      150.00

Ultrasound                   1      200.00

MRI Scan                     1      800.00

Vaccination                  1       60.00



                                                    Healthcare System

Medicine                   Qty Amount (RM)
------------------------- ---- -----------
Paracetamol                  2       10.00

Amoxicillin                  5       60.00

Cough Syrup                  1        8.50

Ibuprofen                    1        7.50

Bandage                     12       36.00



                                                    Healthcare System

Grand Total (RM)
----------------
        3,840.00
```

Purpose:

-Show all invoices, services, and medicines for a specific patient.

-Track the total amount spent by the patient on services and medicines.

-Organize the data by invoice so it's easy to see detailed charges.

Importance：

-Helps hospital staff and management verify patient billing and charges.

-Supports transparency and reduces billing disputes with patients.

-Makes it easier to see which services or medicines contribute most to revenue for each patient.

6.2 Service Used by Patient (Ram)

```
-- Set up the environment for the report
SET PAGESIZE 50
SET LINESIZE 120
SET ECHO OFF

-- Clear any existing column and compute settings
CLEAR COLUMNS
CLEAR COMPUTE
CLEAR BREAKS

-- Prompt the user for the Patient ID
ACCEPT p_patient_id PROMPT 'Enter Patient ID: '

-- Define the report title
TTITLE CENTER 'Services Used by Patient' SKIP 2

-- Define column formatting for readability
COLUMN Patient_Name HEADING 'Patient Name' FORMAT A30
COLUMN Service_Name HEADING 'Service Name' FORMAT A30
COLUMN Service_Price HEADING 'Price (RM)' FORMAT 9999.99
COLUMN Quantity HEADING 'Qty' FORMAT 99
```

```
COLUMN Total_Cost HEADING 'Total (RM)' FORMAT 99999.99

-- Define report breaks and computations
BREAK ON Patient_Name SKIP 1
COMPUTE SUM OF Total_Cost ON Patient_Name

-- The main query to retrieve the service details for the specified patient
SELECT
    p.Patient_Name,
    s.Service_Name,
    s.Service_Price,
    iserv.Quantity,
    (s.Service_Price * iserv.Quantity) AS Total_Cost
FROM
    PATIENT p
JOIN
    INVOICE i ON p.Patient_ID = i.Patient_ID
JOIN
    INVOICE_SERVICE iserv ON i.Invoice_ID = iserv.Invoice_ID
JOIN
    SERVICE s ON iserv.Service_ID = s.Service_ID
WHERE
    p.Patient_ID = &p_patient_id
ORDER BY
    s.Service_Name;
```

```
SQL> start "\\wsl.localhost\Ubuntu\home\ram\Database\Query.txt"
Enter Patient ID: 1001
old  16:     p.Patient_ID = &p_patient_id
new  16:     p.Patient_ID = 1001

                                        Services Used by Patient

Patient Name                  Service Name                   Price (RM) Qty Total (RM)
----------------------------- ------------------------------ ---------- --- ----------
Ali Hassan                    Blood Test                          80.00   1      80.00
                              Consultation                        50.00   1      50.00
                              ECG                                150.00   1     150.00
                              MRI Scan                           800.00   1     800.00
                              Physiotherapy                      100.00   1     100.00
                              Vaccination                         60.00   1      60.00
                              X-Ray                              120.00   1     120.00
**************************                                                   ----------
sum                                                                         1360.00


7 rows selected.

SQL>
```

Purpose:

Used to track how many services were issued to a specific patient, along with the total spending over all visits.

6.3 Invoices by Patient (Eugene Choong Jin Le)

How many invoices were issued to a specific patient, including total spending and services/products charged?

SPOOL invoices_by_patient.txt

TTITLE CENTER 'Invoices by Patient Report'

ACCEPT p_patient_id PROMPT 'Enter Patient ID: '

ACCEPT start_date PROMPT 'Enter Start Date (YYYY-MM-DD): '

ACCEPT end_date PROMPT 'Enter End Date (YYYY-MM-DD): '

COLUMN invoice_id HEADING 'Invoice ID' FORMAT 999999

COLUMN invoice_date HEADING 'Invoice Date' FORMAT A12

```
COLUMN patient_name   HEADING 'Patient Name'    FORMAT A20
COLUMN total_amount   HEADING 'Total Amount (RM)' FORMAT 999,999.99

BREAK ON patient_name SKIP 1
COMPUTE COUNT LABEL 'Total Invoices:' OF invoice_id ON patient_name
COMPUTE SUM   LABEL 'Total Spending (RM):' OF total_amount ON patient_name

SELECT
   i.invoice_id,
   TO_CHAR(i.invoice_date, 'YYYY-MM-DD') AS invoice_date,
   p.patient_name,
   i.total_amount
FROM invoice i
JOIN patient p
   ON i.patient_id = p.patient_id
WHERE i.patient_id = '&p_patient_id'
  AND i.invoice_date BETWEEN TO_DATE('&start_date', 'YYYY-MM-DD')
                         AND TO_DATE('&end_date', 'YYYY-MM-DD')
ORDER BY i.invoice_date;

SPOOL OFF;
```

```
SQL> start C:\Users\choon\Downloads\test.txt
Enter Patient ID: 1002
Enter Start Date (YYYY-MM-DD): 2025 1 1
Enter End Date (YYYY-MM-DD): 2025 12 30
old   9: WHERE i.patient_id = '&p_patient_id'
new   9: WHERE i.patient_id = '1002'
old  10:   AND i.invoice_date BETWEEN TO_DATE('&start_date', 'YYYY-MM-DD')
new  10:   AND i.invoice_date BETWEEN TO_DATE('2025 1 1', 'YYYY-MM-DD')
old  11:                          AND TO_DATE('&end_date', 'YYYY-MM-DD')
new  11:                          AND TO_DATE('2025 12 30', 'YYYY-MM-DD')

                        Invoices by Patient Report
Invoice ID Invoice Date Patient Name         Total Amount (RM)
---------- ------------ -------------------- -----------------
      1522 2025-11-10   Siti Aminah                       60.00
      1502 2025-11-30                                    350.00
---------- ------------ -------------------- -----------------
         2              ********** Total Invoices:
                        Total Spending (RM):              410.00
```

Purpose:

Used to track how many invoices were issued to a specific patient, along with the total healthcare spending over a selected period.

Importance:

- Helps hospital administration monitor patient billing history
- Useful for financial audits, payment disputes, and insurance verification
- Supports decision-making related to high-cost patients and treatment patterns

6.4 Doctor clinical checkup & medicine issued (Wong Yi Hoe)

```
SET PAGESIZE 50

SET LINESIZE 120

SET VERIFY OFF

SET FEEDBACK OFF

CLEAR COLUMNS

CLEAR COMPUTE

CLEAR BREAKS
```

```
TTITLE '================================================================================' SKIP 1 -
      '          DOCTOR CLINICAL CHECKUP and MEDICINE ISSUED REPORT ' SKIP 1 -
      '================================================================================' SKIP 2

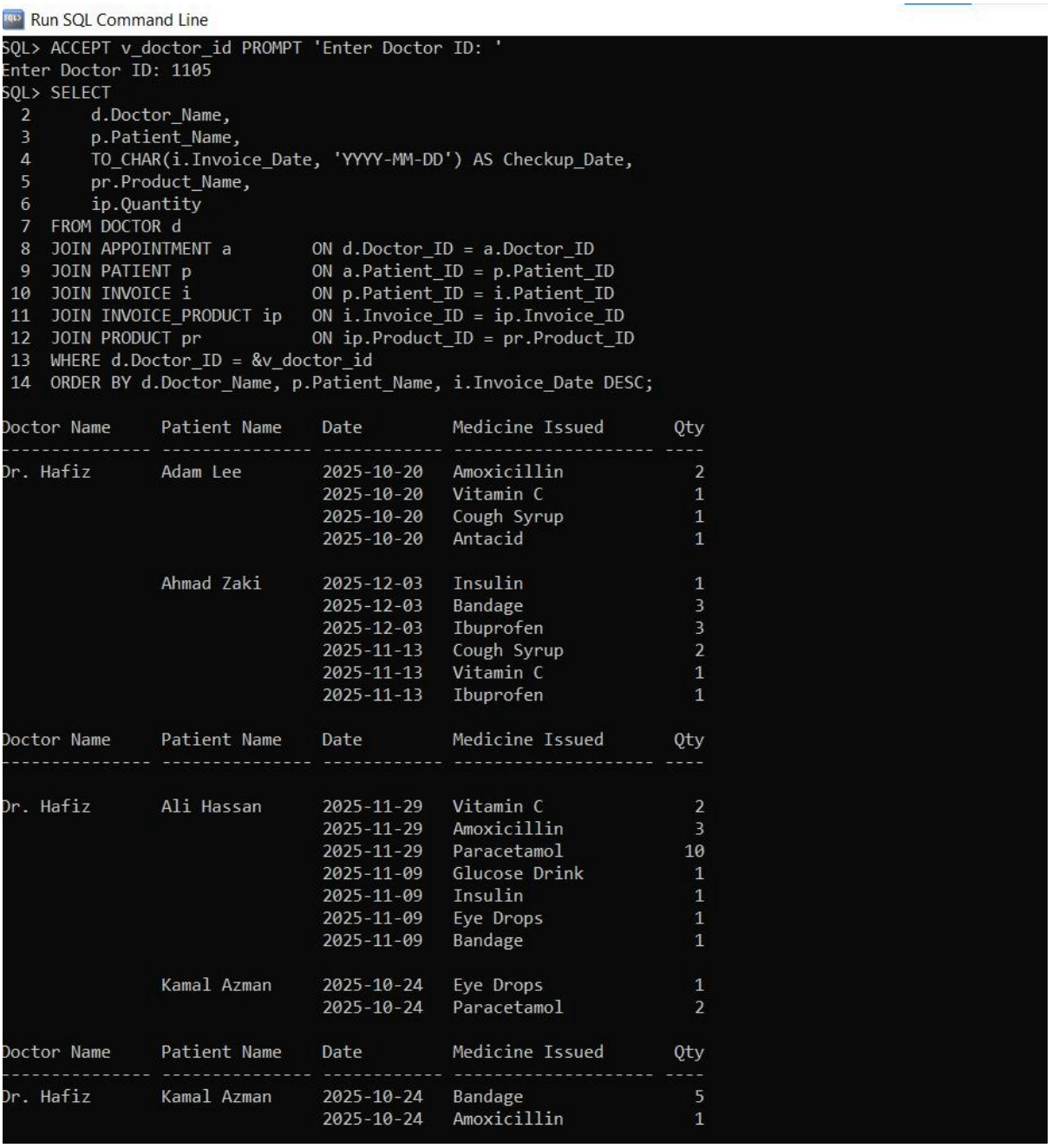
COLUMN Doctor_Name   HEADING 'Doctor Name'     FORMAT A15
COLUMN Patient_Name  HEADING 'Patient Name'    FORMAT A15
COLUMN Checkup_Date  HEADING 'Date'            FORMAT A12
COLUMN Product_Name  HEADING 'Medicine Issued' FORMAT A20
COLUMN Quantity      HEADING 'Qty'             FORMAT 999

BREAK ON Doctor_Name ON Patient_Name SKIP 1

ACCEPT v_doctor_id PROMPT 'Enter Doctor ID: '

SELECT
    d.Doctor_Name,
    p.Patient_Name,
    TO_CHAR(i.Invoice_date, 'YYYY-MM-DD') AS Checkup_Date,
    pr.Product_Name,
    ip.Quantity
FROM DOCTOR d
JOIN APPOINTMENT a        ON d.Doctor_ID = a.Doctor_ID
JOIN PATIENT p            ON a.Patient_ID = p.Patient_ID
JOIN INVOICE i            ON p.Patient_ID = i.Patient_ID
JOIN INVOICE_PRODUCT ip   ON i.Invoice_ID = ip.Invoice_ID
JOIN PRODUCT pr           ON ip.Product_ID = pr.Product_ID
WHERE d.Doctor_ID = &v_doctor_id
ORDER BY d.Doctor_Name, p.Patient_Name, i.Invoice_Date DESC;

TTITLE OFF
SET FEEDBACK ON
```



```
Run SQL Command Line
SQL> ACCEPT v_doctor_id PROMPT 'Enter Doctor ID: '
Enter Doctor ID: 1105
SQL> SELECT
  2      d.Doctor_Name,
  3      p.Patient_Name,
  4      TO_CHAR(i.Invoice_Date, 'YYYY-MM-DD') AS Checkup_Date,
  5      pr.Product_Name,
  6      ip.Quantity
  7  FROM DOCTOR d
  8  JOIN APPOINTMENT a        ON d.Doctor_ID = a.Doctor_ID
  9  JOIN PATIENT p            ON a.Patient_ID = p.Patient_ID
 10  JOIN INVOICE i            ON p.Patient_ID = i.Patient_ID
 11  JOIN INVOICE_PRODUCT ip   ON i.Invoice_ID = ip.Invoice_ID
 12  JOIN PRODUCT pr           ON ip.Product_ID = pr.Product_ID
 13  WHERE d.Doctor_ID = &v_doctor_id
 14  ORDER BY d.Doctor_Name, p.Patient_Name, i.Invoice_Date DESC;

Doctor Name     Patient Name    Date         Medicine Issued       Qty
--------------- --------------- ------------ -------------------- ----
Dr. Hafiz       Adam Lee        2025-10-20   Amoxicillin             2
                                2025-10-20   Vitamin C               1
                                2025-10-20   Cough Syrup             1
                                2025-10-20   Antacid                 1

                Ahmad Zaki      2025-12-03   Insulin                 1
                                2025-12-03   Bandage                 3
                                2025-12-03   Ibuprofen               3
                                2025-11-13   Cough Syrup             2
                                2025-11-13   Vitamin C               1
                                2025-11-13   Ibuprofen               1

Doctor Name     Patient Name    Date         Medicine Issued       Qty
--------------- --------------- ------------ -------------------- ----

Dr. Hafiz       Ali Hassan      2025-11-29   Vitamin C               2
                                2025-11-29   Amoxicillin             3
                                2025-11-29   Paracetamol            10
                                2025-11-09   Glucose Drink           1
                                2025-11-09   Insulin                 1
                                2025-11-09   Eye Drops               1
                                2025-11-09   Bandage                 1

                Kamal Azman     2025-10-24   Eye Drops               1
                                2025-10-24   Paracetamol             2

Doctor Name     Patient Name    Date         Medicine Issued       Qty
--------------- --------------- ------------ -------------------- ----
Dr. Hafiz       Kamal Azman     2025-10-24   Bandage                 5
                                2025-10-24   Amoxicillin             1
```

Purpose:

-Detailed Patient File Review: This report is specifically designed for healthcare staff to review a patient's medical file, allowing them to see exactly which medicines were issued and by which doctor.

-Organized Historical Documentation: By grouping data by doctor and patient name, the report creates a chronological log of checkups, making it easy to see the progression of treatment and the medicine quantities provided over time

Importance:

-Hospital Administrative Monitoring: The report provides management with a clear window into billing history, ensuring that every medical item issued is properly recorded and accounted for in the system.

-Support for Resource Management: It helps hospital administrators identify high-cost patients and analyze treatment patterns, which is vital for making informed decisions about pharmacy stock and hospital financial planning.

6.5 Total appointments accepted by each Department

Run SQL Command Line

```
  8  JOIN
  9      APPOINTMENT a ON d.Doctor_ID = a.Doctor_ID
 10  JOIN
 11      INVOICE i ON a.Patient_ID = i.Patient_ID
 12  JOIN
 13      INVOICE_SERVICE iser ON i.Invoice_ID = iser.Invoice_ID
 14  JOIN
 15      SERVICE s ON iser.Service_ID = s.Service_ID
 16  WHERE
 17      a.Appointment_Date BETWEEN TO_DATE('&v_start_date', 'YYYY-MM-DD')
 18                             AND TO_DATE('&v_end_date', 'YYYY-MM-DD')
 19  GROUP BY
 20      d.Doctor_Name, d.Specialization
 21  ORDER BY
 22      "Revenue Generated" DESC;
old  17:     a.Appointment_Date BETWEEN TO_DATE('&v_start_date', 'YYYY-MM-DD')
new  17:     a.Appointment_Date BETWEEN TO_DATE('2025-11-11', 'YYYY-MM-DD')
old  18:                             AND TO_DATE('&v_end_date', 'YYYY-MM-DD')
new  18:                             AND TO_DATE('2025-12-19', 'YYYY-MM-DD')

================================================
DOCTOR APPOINTMENT LOAD AND REVENUE ANALYSIS REPORT
Reporting Period: 2025-11-11 to 2025-12-19
================================================

Doctor Name              SPECIALIZATION       Total Appointments Revenue Generated
------------------------ -------------------- ------------------ -----------------
Dr. Ahmad                General                               2         $2,660.00
Dr. Sarah                Psychiatry                            2         $2,640.00
Dr. Aina                 Gynecology                            2         $2,560.00
Dr. Hafiz                Orthopedic                            2         $2,450.00
Dr. Siti                 Pediatrics                            2         $2,380.00
Dr. Wong                 Dermatology                           2         $1,900.00
Dr. Daniel               Ophthalmology                         2         $1,800.00
Dr. Lim                  Cardiology                            2         $1,740.00
Dr. John                 ENT                                   2         $1,690.00
Dr. Kumar                Neurology                             1           $630.00
                                                                      -----------------
sum                                                                      $20,450.00

10 rows selected.

SQL>
SQL> TTITLE OFF
SQL> CLEAR BREAK
breaks cleared
SQL> CLEAR COMPUTE
computes cleared
SQL> SET FEEDBACK ON
SQL> SET VERIFY ON
```

SET LINESIZE 200

SET PAGESIZE 100

SET FEEDBACK OFF

SET VERIFY OFF

```
ACCEPT v_start_date PROMPT 'Enter Start Date (YYYY-MM-DD): '
ACCEPT v_end_date   PROMPT 'Enter End Date (YYYY-MM-DD): '

COLUMN "Doctor Name" FORMAT A25
COLUMN "Specialization" FORMAT A20
COLUMN "Total Appointments" FORMAT 999
COLUMN "Revenue Generated" FORMAT $99,999.90

TTITLE  '================================================' SKIP 1
-
      'DOCTOR APPOINTMENT LOAD AND REVENUE ANALYSIS REPORT' SKIP
1 -
      'Reporting Period: &v_start_date to &v_end_date' SKIP 1 -
      '================================================' SKIP 2

BREAK ON REPORT
COMPUTE SUM OF "Revenue Generated" ON REPORT
SELECT
   d.Doctor_Name AS "Doctor Name",
   d.Specialization,
   COUNT(DISTINCT a.Appointment_ID) AS "Total Appointments",
   SUM(s.Service_Price * iser.Quantity) AS "Revenue Generated"
FROM
   DOCTOR d
JOIN
   APPOINTMENT a ON d.Doctor_ID = a.Doctor_ID
JOIN
   INVOICE i ON a.Patient_ID = i.Patient_ID
```

```
JOIN
   INVOICE_SERVICE iser ON i.Invoice_ID = iser.Invoice_ID
JOIN
   SERVICE s ON iser.Service_ID = s.Service_ID
WHERE
   a.Appointment_Date BETWEEN TO_DATE('&v_start_date', 'YYYY-MM-DD')
                          AND TO_DATE('&v_end_date', 'YYYY-MM-DD')
GROUP BY
   d.Doctor_Name, d.Specialization
ORDER BY
   "Revenue Generated" DESC;

TTITLE OFF
CLEAR BREAK
CLEAR COMPUTE
SET FEEDBACK ON
SET VERIFY ON
```

Purpose：

1.Show how many appointments each department has in a certain period.

2.Track revenue generated from the services each department provides.

3.Organize the data by doctor so it’s easy to see trends and workload.

Importance：

1.Helps hospital management check department workloads and earnings.

2.Supports planning for staff schedules and hospital finances.

3.Makes it easier to see which department handle more patients or generate more revenue.